Hovestadt 2019 medulloblastoma — analysis summary

GO / NO-GO GATES

Gate 1 — mixing & continuity → GO

25 samples intermix per-neighbourhood on one continuous manifold (iLISI 1.17→7.67; 0 islands removed). Judged on the regenerated embedding across a Harmony {2,5,10} sweep — no per-sample archipelago despite strong medulloblastoma molecular-subgroup batch effect. Cycling compartment present but Gast is not proliferation-confounded (r=−0.13), so the embryonic axis is real and the whole manifold is retained.

Gate 2 — module geography → GO (non-branching)

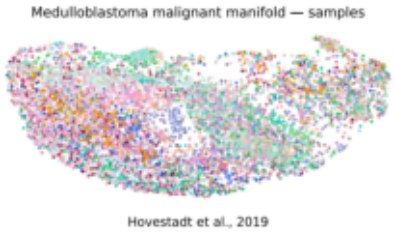
Non-branching case: Gast & pSom co-vary (r=+0.46, overlap — no two competing arms), Crest an inverse terminal gradient (Crest–Gast r=−0.90, Crest–pSom r=−0.67). Single (Gast/pSom)→Crest developmental axis; Crest-high decile at pseudotime 0.82 (terminal). Passes Gate 2 as the non-branching outcome.

Gate 3 — cross-repression → NA

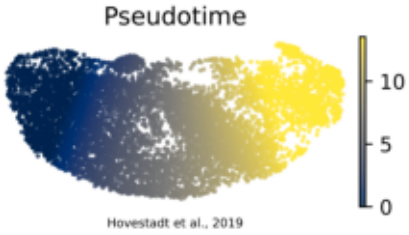
No bifurcation — no hub to test. Gate 3 is branching-only, so it is Not Applicable for this linear manifold.

STEP FIGURES

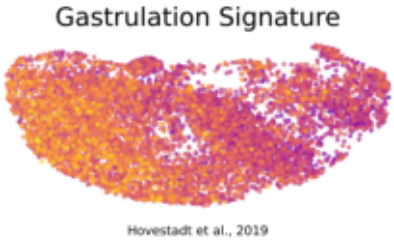
Manifold — samples



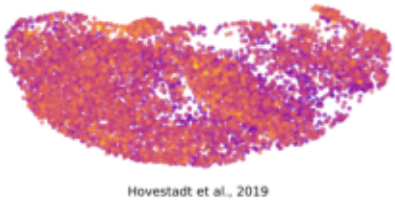
Pseudotime



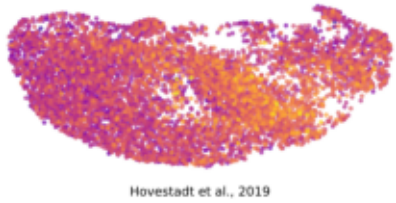
Gastrulation



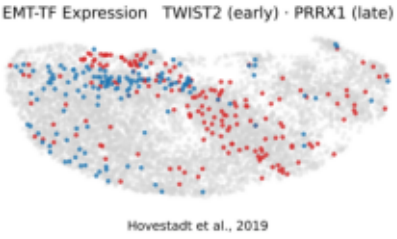
Somitogenesis



Neural Crest



EMT-TF expression



Dataset.

Medulloblastoma (Hovestadt et al., 2019), SmartSeq2 — 7,921 malignant cells across 25 samples (of 36; ≥100-cell floor). Non-branching Linear-pSom/Crest.

Verdict.

Pattern reproduces as the non-branching case — one embryonic Gast/pSom → Crest axis, Crest inverse-terminal; no bifurcation (Gate 3 NA). Rooted at the Gast/pSom-high pole (Gast early, Crest late); the embryonic axis is not proliferation-confounded so the whole manifold is retained. EMT-TFs TWIST2 (early/root) → PRRX1 (late/terminus).

Parameter choices

Min-cell threshold	≥100 (no cliff) → 25/36 samples, 7,921 cells	**Integration**	Harmony on sample θ=5, n_pcs=20 (mixing saturates 5–10)
Island removal	0 cells	**Geometry**	Linear (non-branching); scFates curve N=20, μ=100, fold-free (2.97)
Root node	curve tip 2 = Gast/pSom-high, Crest-low origin	**Module means**	Gast −0.043 · pSom +0.186 · Crest −0.026
EMT-TFs	TWIST2 (early, pt 0.24) → PRRX1 (late/terminus, pt 0.50)		